

		Potency	Starting Dose (mg)	Target Dose Range (mg/day)	Motor Side Effects	Endocrine (Prolactinemia)	Hypotension	Anticholinergic	Sedation	Weight	Metabolic Dysfunction	Miscellaneous	Dosage Strengths	Other Formulations
First Generation Antipsychotics (FGAs)	Chlorpromazine (Thorazine)	↓	50 BID	400–1000	↓	+++	+++	+++	Hi	+++	+	Metabolite is amoxapine (antidepressant)	10, 25, 50, 100, 200 mg tablets	25 mg/mL inj.
	Loxapine (Loxitane)		20 BID	60–100		++	++	+	↑	±↓	±		5, 10, 25, 50 mg capsules	10 mg inhal. powder
	Perphenazine (Trilafon)		4–8 BID	24–48		++	+	+		++	±		2, 4, 8, 16 mg tablets	—
	Thiothixene (Navane)		5–10 QD	10–60		++	+	+		±	±	Rare lens pigmentation, long ½ life	1, 2, 5, 10 mg capsules	—
	Trifluoperazine (Stelazine)		2–5 BID	10–30		++	+	+		+	±		1, 2, 5, 10 mg tablets	—
	Fluphenazine (Prolixin)		2–5 BID	5–20		+++	+	+		+	±		1, 2.5, 5, 10 mg tablets	5 mg/mL oral sol, 2.5 mg/mL inj. & 25 mg/mL depot
	Haloperidol (Haldol)		2–5 BID	5–20		+++	+	+		+	±	Depot: 100–200 mg q 4 weeks	0.5, 1, 2, 5, 10, 20 mg tablets	2 mg/mL oral sol, 5 mg/mL inj. & 50, 100 mg/mL depot
	Pimozide (Orap)		1–2 QD	2–8		+++	+	+	Lo	+	±		1, 2 mg tablets	—
												↑QTc—monitor EKG and K ⁺ , CYP3A & 1A2, long ½ life		
Second-Generation Antipsychotics (SGAs)	Quetiapine (Seroquel)	↓	50–100 BID	400–1000	↔	↔	++	+	+++	++	++	SE: activation, insomnia, Should be taken WITH food Dose dependent increase in SE, should be taken WITH food. CYP3A4 interaction On Formulary only for Bipolar Depression LAI: 210 or 300 mg q 2 weeks, 405 mg q 4 weeks	25, 50, 100, 200, 300, 400 mg tablets	50, 150, 200, 300, 400 mg extended release tablets
	Ziprasidone (Geodon)		20 BID	160–320	+	↔	+	↔	↔	↔	↔		20, 40, 60, 80 mg capsules	20 mg inj.
	Lurasidone (Latuda)		40 QD	40–80	↔	↔	+	↔	++	↔	↔		20, 40, 60, 80, 120 mg tablets	—
	Olanzapine (Zyprexa)		5–10 QD	10–40	±	↔	+	+	++	+++	+++	SE: activation, insomnia, nausea, vomiting LAI: 400 mg q 4 weeks SE: dizziness, somnolence, dyspepsia, dry mouth. Gradually titrate over 1 week and dose BID to avoid orthostatic hypotension. Kit available to determine CYP2D6 genotype. SE: headache, tachycardia Renal metabolism Sustenna q 4 weeks IM injection, start with 234 mg and then 156 mg 1 week later SE: oral hypoesthesia, constipation, nausea and vomiting, insomnia ^o consta: 25–50 mg q 2 weeks	2.5, 5, 7.5, 10, 15, 20 mg tablets	10 mg inj., 210, 300, 405 mg LAI., 5, 10, 15, 20 mg ODT
	Aripiprazole (Abilify)		10–15 QD	10–45	+	↔	+	↔	↔	↔	↔		2, 5, 10, 15, 20, 30 mg tablets	300, 400 mg LAI, 9.75 mg/1.3 mL inj., & 10, 15 mg ODT, 1mg/mL oral sol
	Iloperidone (Fanapt)		1 BID titrated slowly	6–12 BID	↔	↔	+++	↔	++	+	↔		1, 2, 4, 6, 8, 10, 12 mg tablets	Titration pack available
	Paliperidone (Invega)		3–6 QD	12	+	+++	+	↔	+	++	++	SE: headache, tachycardia Renal metabolism Sustenna q 4 weeks IM injection, start with 234 mg and then 156 mg 1 week later	1.5, 3, 6, 9 mg tablets	39, 78, 117, 156, 234 mg LAI
	Asenapine (Saphris)		5 BID	5–10	+	↔	+	↔	+	+	↔	SE: oral hypoesthesia, constipation, nausea and vomiting, insomnia	5, 10 mg SL tablets	—
	Risperidone (Risperdal)		1–2 QD	4–8	+	+++	+	↔	+	++	++	^o consta: 25–50 mg q 2 weeks	0.25, 0.5, 1, 2, 3, 4 mg tablets	1 mg/mL oral sol & 12.5, 25, 37.5, 50 mg inj., ODT
	Clozapine (Clozaril)	Lo	25 QD	300–600	↔	↔	+++	+++	+++	+++	+++	SE: agranulocytosis, seizures, myocarditis, cardiomyopathy Weekly CBC x 6 months → q 2 weeks x 6 months → q month	25, 50, 100, 200 mg tablets	12.5, 25, 100 mg ODT, 50 mg/mL oral susp

QD = daily, BID = twice daily, SE = side effects, ODT = orally disintegrating tablets, LAI = long acting injection, SL = sublingual.

1. SIDE EFFECTS PROFILE

Relative risk of EPS: High-potency FGAs > mid-potency FGAs = risperidone = paliperidone > low-potency FGA > aripiprazole ≥ ziprasidone = asenapine = olanzapine > lurasidone > iloperidone > quetiapine > clozapine

Relative risk of TD: FGA > SGA > clozapine

Relative risk of weight gain: clozapine = olanzapine > low-potency FGA > risperidone = paliperidone = quetiapine > iloperidone > asenapine > mid-potency FGA > high-potency FGA = aripiprazole = ziprasidone = lurasidone

Relative risk of prolactin elevation and sexual side effects: risperidone = paliperidone > FGA > olanzapine > ziprasidone > quetiapine = clozapine > aripiprazole > asenapine > lurasidone > iloperidone

2. EFFICACY: No significant differences in efficacy—except clozapine

3. FIRST EPISODE: Any FGA or SGA—choice driven by side effects. Use lower dose of FGA (half?) to reduce the risk of EPS.

Quicker response to positive symptoms, more motor side effects and poor adherence is usually seen, so closer monitoring is recommended.

4. CLOZAPINE USE: After failure of 2 antipsychotic trials. Can be the initial choice in patients with suicidality, violence, and substance use. In patients with 2 years of positive symptoms on medication treatment and 5 years of inadequate response, clozapine should be considered.

Metabolic Syndrome		
Definitions: Adult treatment panel III (ATPIII)—presence of 3 or more of the following, International Diabetes Federation (IDF)—central obesity plus 2 of the following:		Monitoring Consensus guidelines by the APA and ADA*
WAIST CIRCUMFERENCE	Male ≥102 cm, Female ≥88 cm (ATPIII) Male ≥94 cm, Female ≥80 cm (IDF)	Baseline and then annually
BP	SBP ≥130 mmHg or DBP ≥85 mmHg or on drug treatment for elevated blood pressure	Baseline, at 12 weeks, and then annually
HDL-C	Male <40 mg/dL, Female <50 mg/dL or on drug treatment for reduced HDL-C	Fasting lipid profile at baseline, then at 12 weeks and then every 5 years
TGL	≥150 mg/dL (1.7 mmol/L) or on drug treatment for raised triglycerides	Fasting lipid profile at baseline, then at 12 weeks and then every 5 years
GLUCOSE	Fasting plasma glucose ≥100 mg/dL (5.6 mmol/L) or on drug treatment for elevated glucose	Baseline, at 12 weeks and then annually

*More frequent assessments of BMI (every 4 weeks for 12 weeks and then quarterly); also monitoring for family history of diabetes, dyslipidemia, and hypertension or cardiovascular diseases is recommended.

SOME SIDE EFFECTS COMMON TO MANY OF THE DRUGS:

Blockade of D₂ dopamine receptors:

Motor side effects: dystonia, akathisia, tremor, rigidity, bradykinesia, akinesia, tardive dyskinesia

Endocrinopathy: gynecomastia, galactorrhea, amenorrhea, sexual dysfunction, decreased libido

Neuroleptic malignant syndrome

Blockade of H₁ histaminic receptors: sedation, weight gain

Blockade of muscarinic receptors: dry mouth, constipation, blurred vision, urinary retention, sinus tachycardia, amnesia

Blockade of α₁ adrenergic receptors: hypotension, dizziness, syncope, reflex tachycardia, priapism

Most drugs can be administered once per day.

Adapted from:

TMAP = The Texas Medication Algorithm Project (Moore TA, et al. The Texas Medication Algorithm Project antipsychotic algorithm for schizophrenia. *J Clin Psychiatry*. 2007;68(11):1751–1762).

PORT = The Schizophrenia Patient Outcomes Research Team (Kreyenbuhl J et al. The Schizophrenia Patient Outcomes Research Team [PORT]: Updated Treatment Recommendations 2009.

Schizophr Bull. 2010;36(1):94–103). © Copyright 2004–2006 International Psychopharmacology Algorithm Project (IPAP) www.ipap.org